Variant to gene to function

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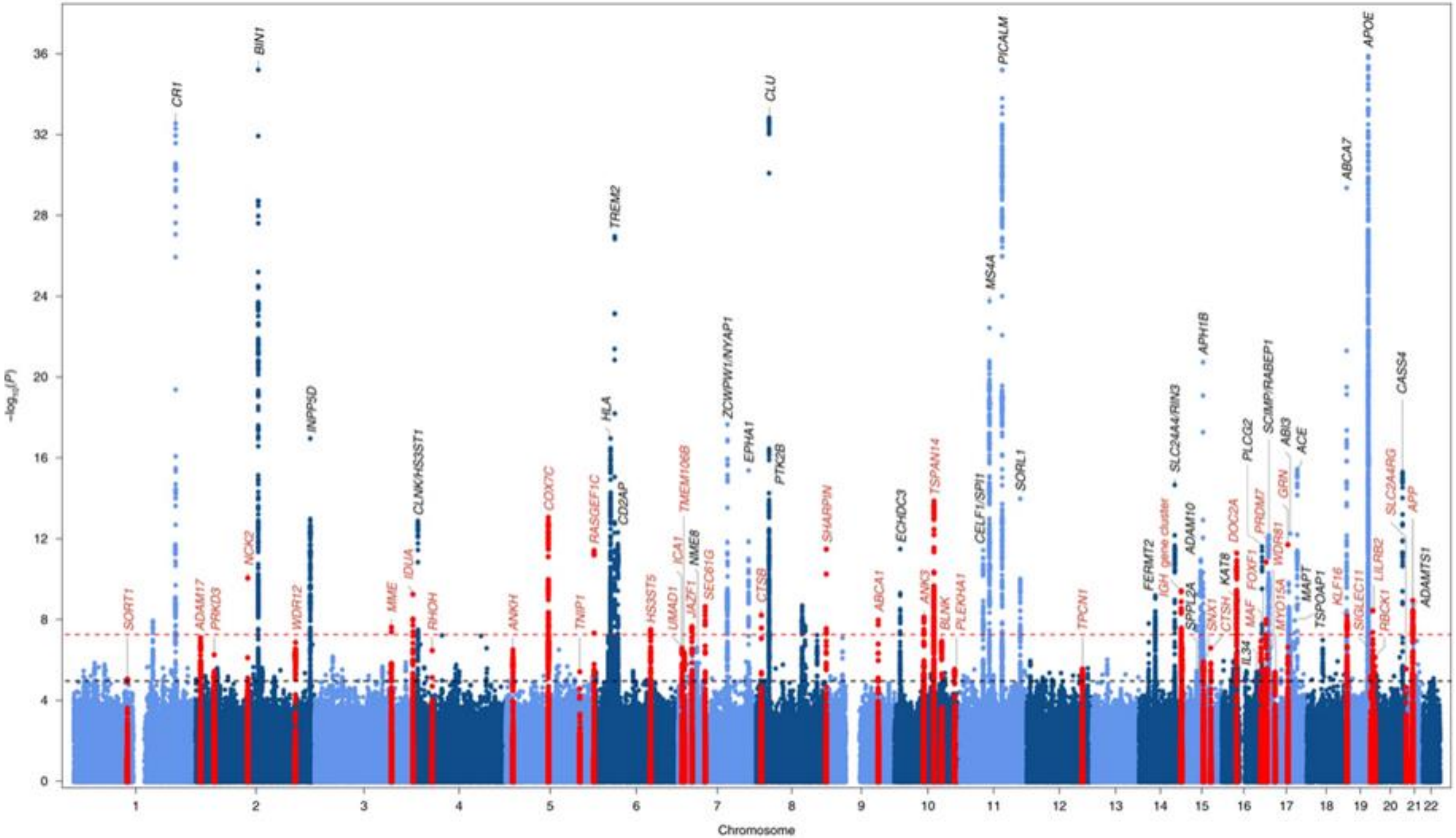
Slides: Gerard Bouland

<u>**G**</u>enome-<u>**W**</u>ide <u>**A**</u>ssociation <u>**S**</u>tudies (GWAS)

Manhattan plot



ref: New insights into the genetic etiology of Alzheimer's disease and related dementias (<u>https://www.nature.com/articles/s41588-022-01024-z</u>)

How does a genome-wide association study find variants?

rs12548

AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**A**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**A**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**A**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**T**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**A**AGAGGAGTCATT

Genomes of individuals with Alzheimer's

AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**T**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**T**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**T**AGAGGAGTCATT
AAACCTCAGCCTGCTTCTCGTCCGGGTTGT**A**AGAGGAGTCATT

Genomes of healthy individuals

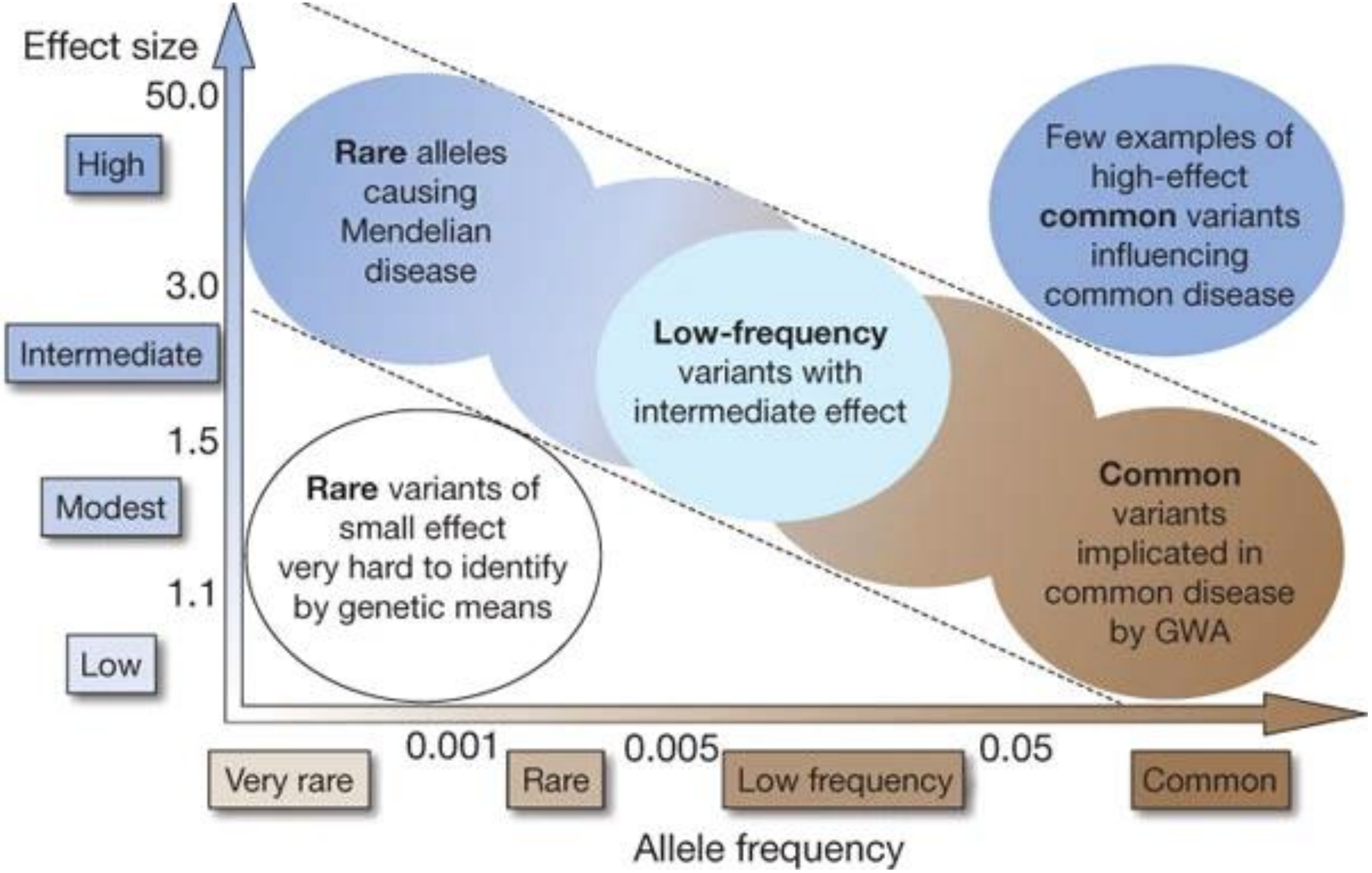
Chi-sq = 25.043
P = 5.61×10^{-7}

Observed	Allele A (N = 50)	Allele T (N = 40)
Alzheimer's (n =50)	40	10
Healthy (n = 40)	10	30

Expected	Allele A (N = 50)	Allele T (N = 40)
Alzheimer's (n =50)	27.8	22.2
Healthy (n = 40)	22.2	17.8

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Most common variants have small effects

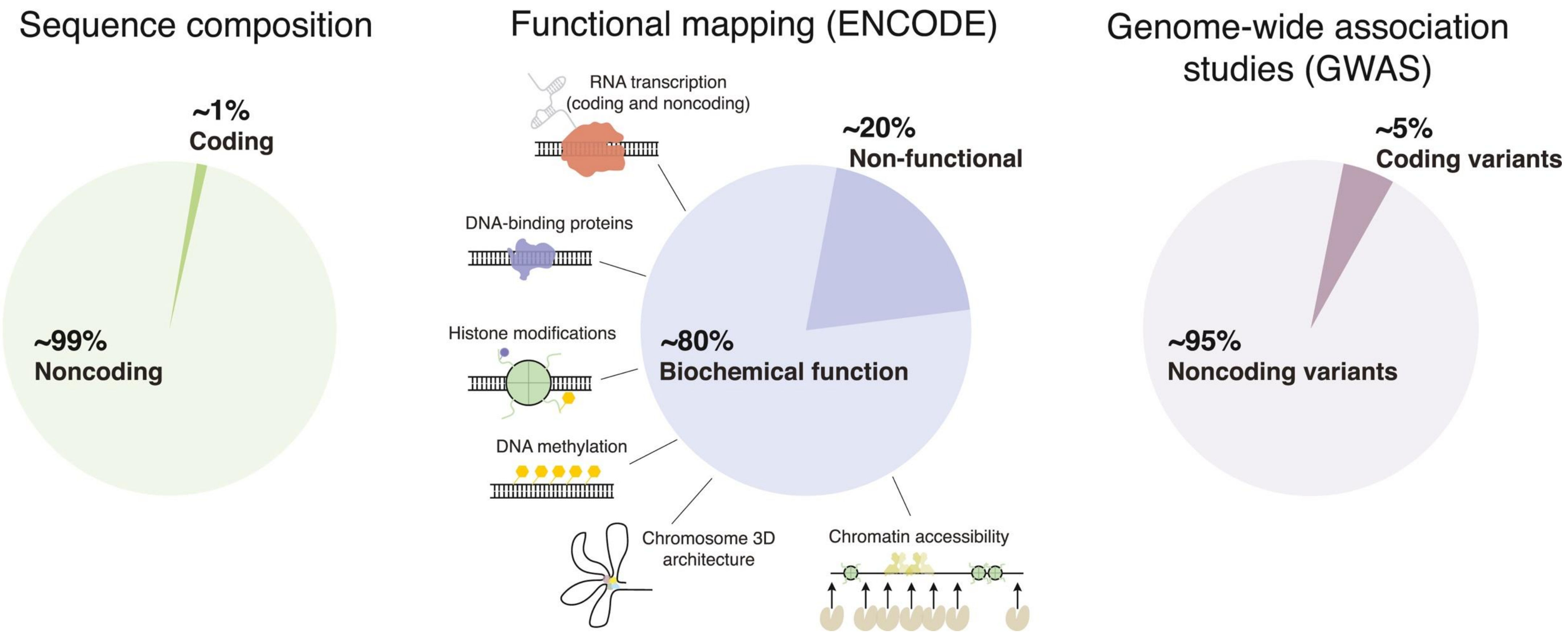


Manolio et al 2009, PMID: 19812666

Disease associated variants

- Alzheimer’s (~ 80 variants)
- Schizophrenia (~290 variants)
- Type 2 Diabetes (~240 variants)
- Chronic Kidney Disease (~40 variants)
- Bipolar disorder (~ 65 variants)

Majority of variants in non-coding regions

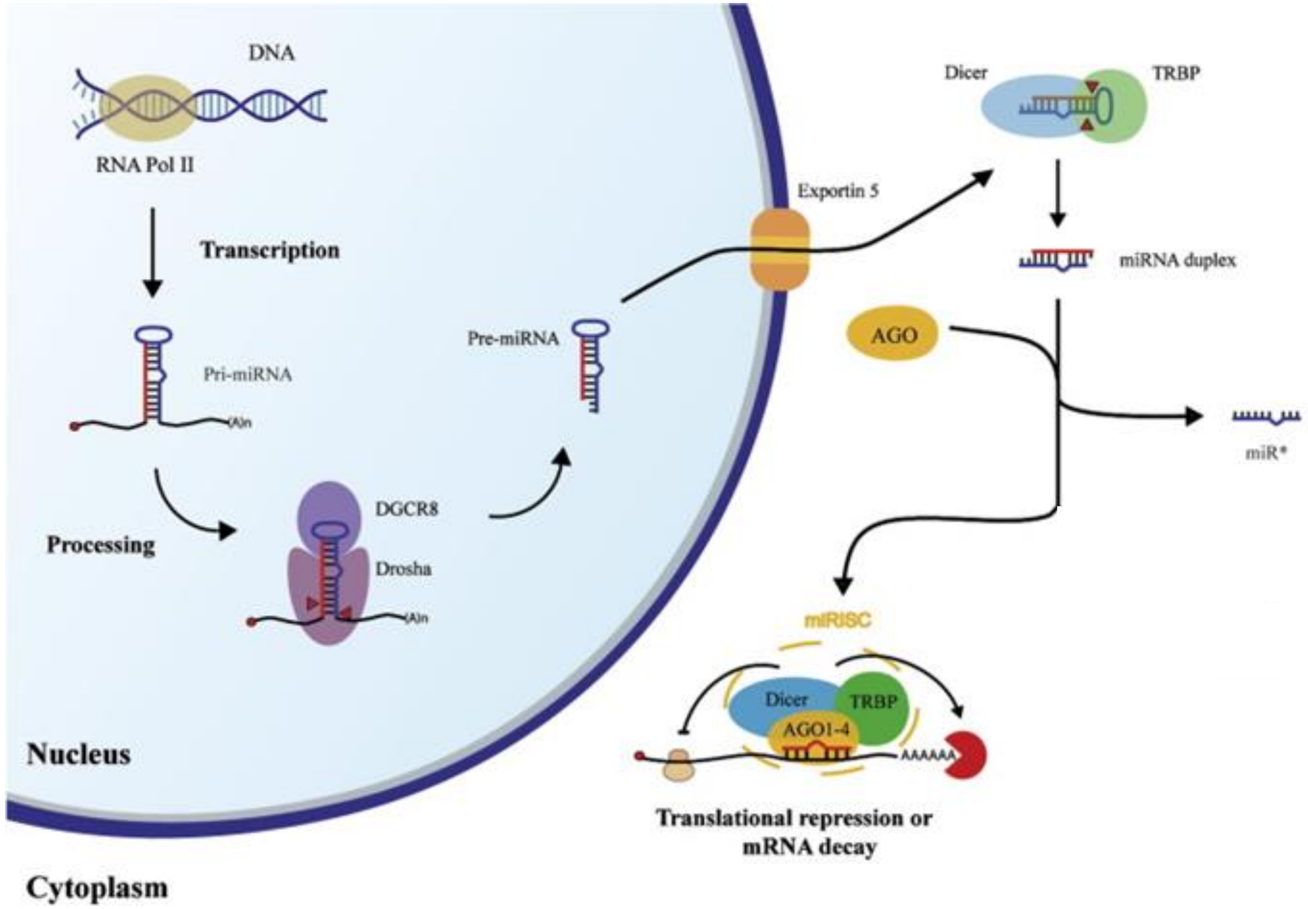


Montalbano et al. *Molecular Cell* 2017

Biological mechanisms of non-coding variants

1. Variant in 3’-UTR region, altering microRNA binding site
2. Variants altering splice sites
3. Variants altering transcription factor binding site
4. Variants altering DNA methylation

Variant in 3'-UTR region, altering microRNA binding site



Saraiva et al, 2017

	Predicted consequential pairing of target region (top) and miRNA (bottom)
Position 21-28 of HMGA2 3' UTR hsa-let-7f-5p	5' ...GCCAACGUUCGAUUUCUACCUCA... ||||||| 3' UUGAUAUGUUAGAUGAUGGAGU

https://www.targetscan.org

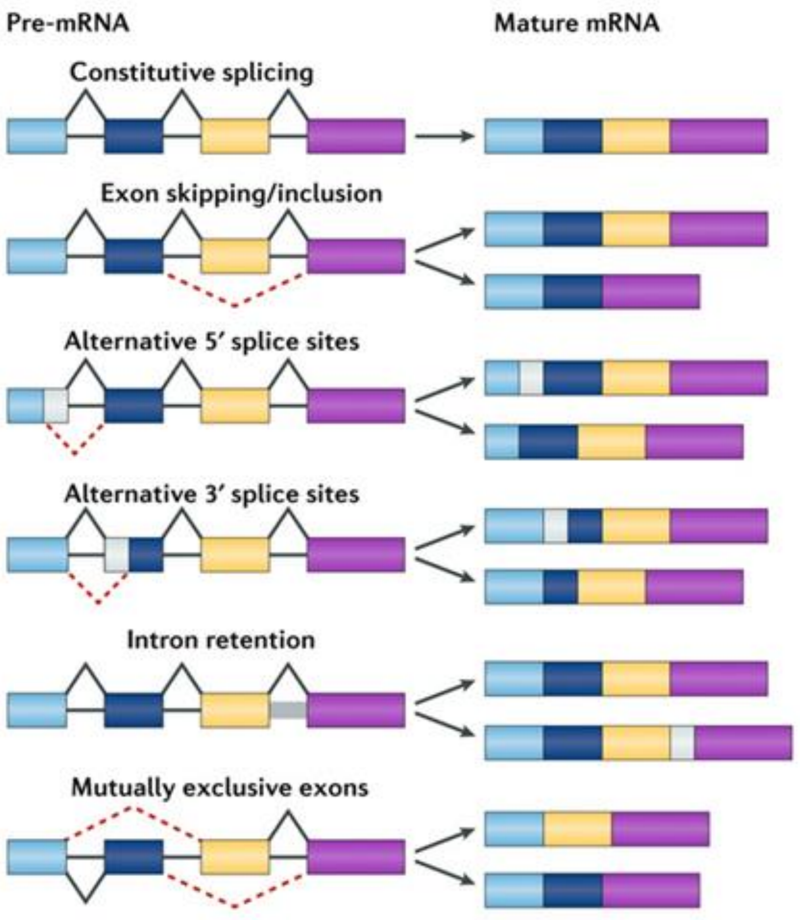
Variant in 3'-UTR region, altering microRNA binding site

- Alzheimer's Disease
  - 10 known variants (Ghanbari et al, 2016)

- Autism
  - 21 known variants (Vaishnavi et al, 2014)

- Metastasis in osteosarcoma
  - 1 known variant (Zhang S et al, 2016)

Variants altering splice sites



Frankiw et al, 2019

- >99% splice sites GT and AG in the donor and acceptor sites, respectively. (Kurmangaliyev et al, 2013)
- Most disease-causing mutations of splice sites occur at these dinucleotides (Kurmangaliyev et al, 2013)

Variants altering splice sites

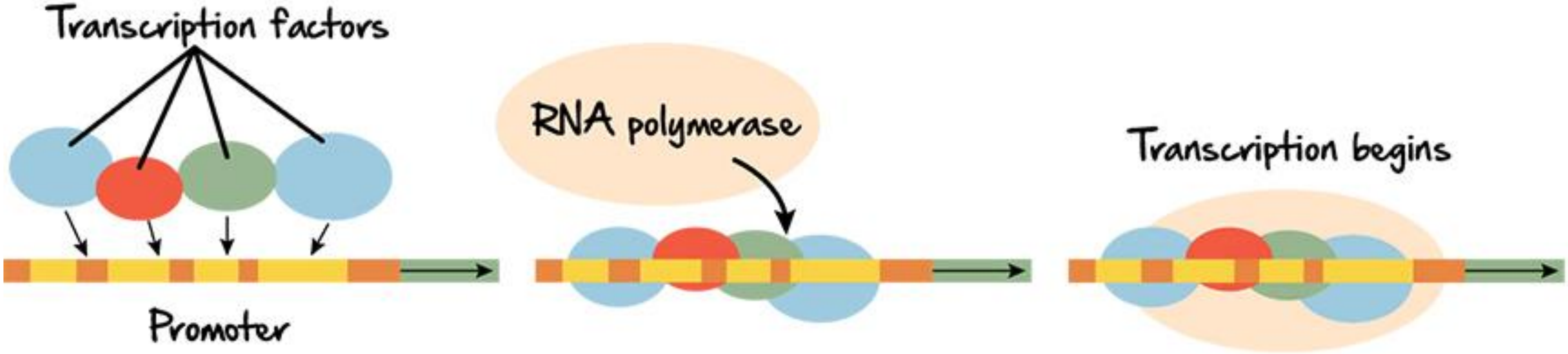
- Adipose-related traits
  - Seven exon-exon junctions in four genes (*AKTIP*, *DTNBP1*, *FTO* and *UBE2E1*) (**Ma et al, 2015**)

- Alzheimer's Disease
  - 21 genes (**Raj, et al, 2018**)

- Schizophrenia
  - Four genes (*NEK4*, *FXR1*, *SNAP91* or *APOPT1)(**Takata et al, 2017**)*

- Breast Cancer
  - Six genes (*BABAM1, DCLRE1B/PHTF1, PEX14, RAD51L1, SRGAP2D and STXBP4*) (**Caswell et al, 2015**)

Variants altering transcription factor binding site



Variants altering transcription factor binding site

- 8% of all polymorphisms reside in TFBS, yet it is estimated that of all disease associated polymorphisms 31% reside in TFBS

- BMI
  - *ARID5B*, *IRX3, IRX5 (***Claussnitzer et al, 2015**)
- Chronic kidney disease
  - *TCF7L2 (***Köttgen et al, 2009**)
- Diabetes
  - *NEUROD1*, *MTNR1B (***Lyssenko et al, 2009**)
- Hypertension
  - *PHOX2*, *SCG2(****Wen et al, 2007***)
- Many many more

Variants altering DNA methylation

- Methylation most often happens at CpG sites

- A methylated cytosine represses gene expression and can even lead to a more condensed chromatin structure.

- 23% of all polymorphisms are CpG site related SNPs (**Zhou et al, 2015**)

- Cardiovascular disease (**Jiantao Ma et al 2022**)
    - *APOB*
    - *SREBF1*
    - *PM20D1*

Summary biological mechanisms of non-coding variants

1. Variant in 3’-UTR region, altering microRNA binding site

2. Variants altering splice sites

3. Variants altering transcription factor binding site

4. Variants altering DNA methylation

Variant to gene

Different ways to link variants to genes

1. Expression quantitative train loci (eQTL)

2. Chromatin interaction

Finding expression quantitative trait loci (eQTLs)

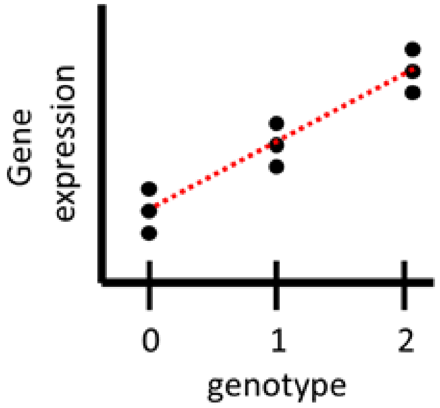
- Expression quantitative trait loci (eQTL) is a SNP that is associated with the expression of a gene.

- An eQTL always forms a pair with an eGene.

- eQTLs are usually identified using linear models.

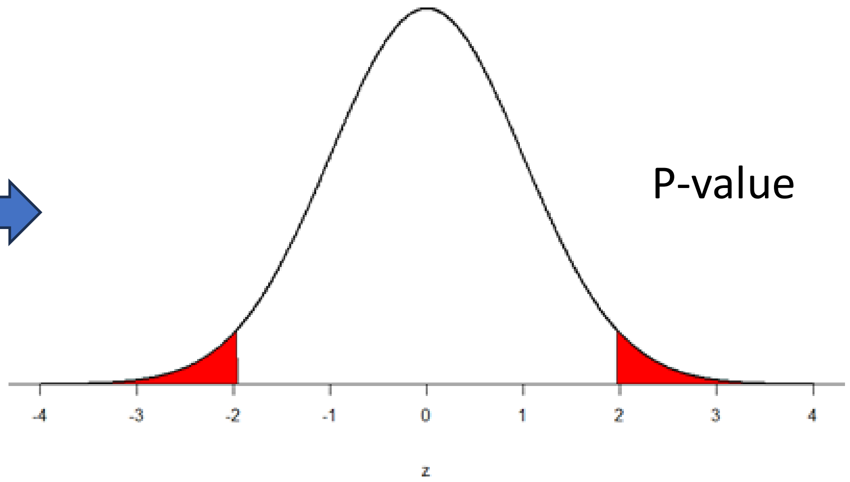
Finding expression quantitative trait loci (eQTLs)

1. Encode genotypes as 0,1,2
   - A/A = 0
   - A/T = 1
   - T/T = 2
2. Associate genotype with gene expression (often linear model)
   - expression = βG + e



$$Z = \frac{\beta}{\text{Standard Error of } \beta}$$

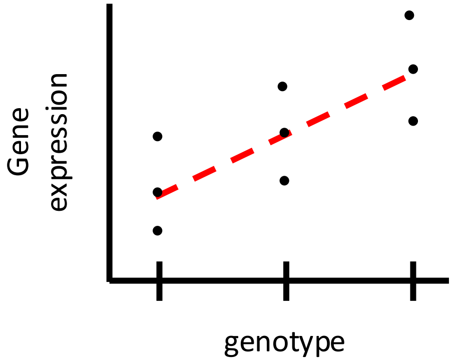
Wald test where,
H0 : β = 0



P-value

Finding expression quantitative trait loci (eQTLs)

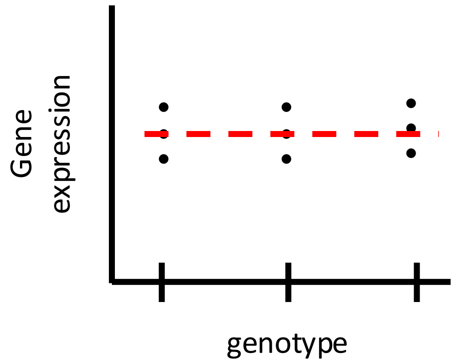
$$Z = \frac{\beta}{\text{Standard Error of } \beta}$$



$\beta = 1$
SE $\beta = 0.33$

Z: 1 / 0.33 = **3**

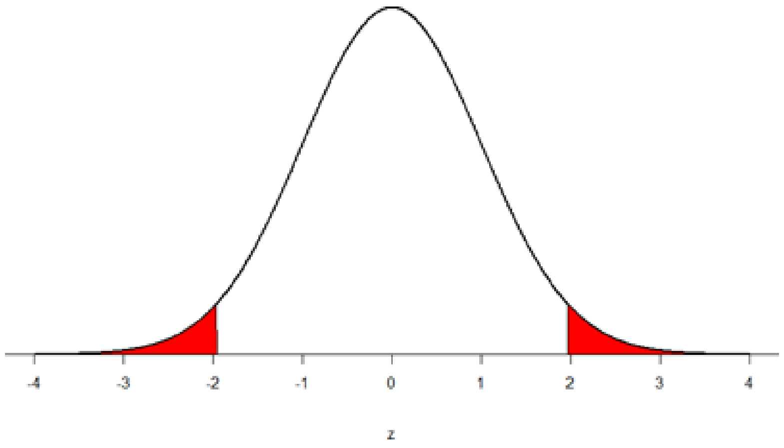
Genotype seems to be predictive of gene expression, but there is still some variability. Small p-value

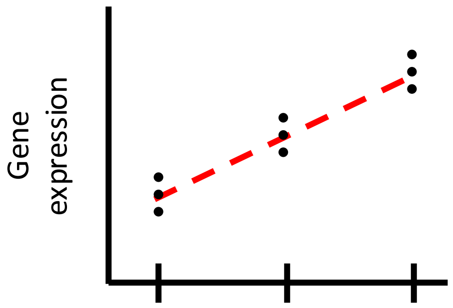


$\beta = 0$
SE $\beta = 0.33$

Z: 0 / 0.33 = **0**

Genotype says nothing about gene expression. P-value = 1





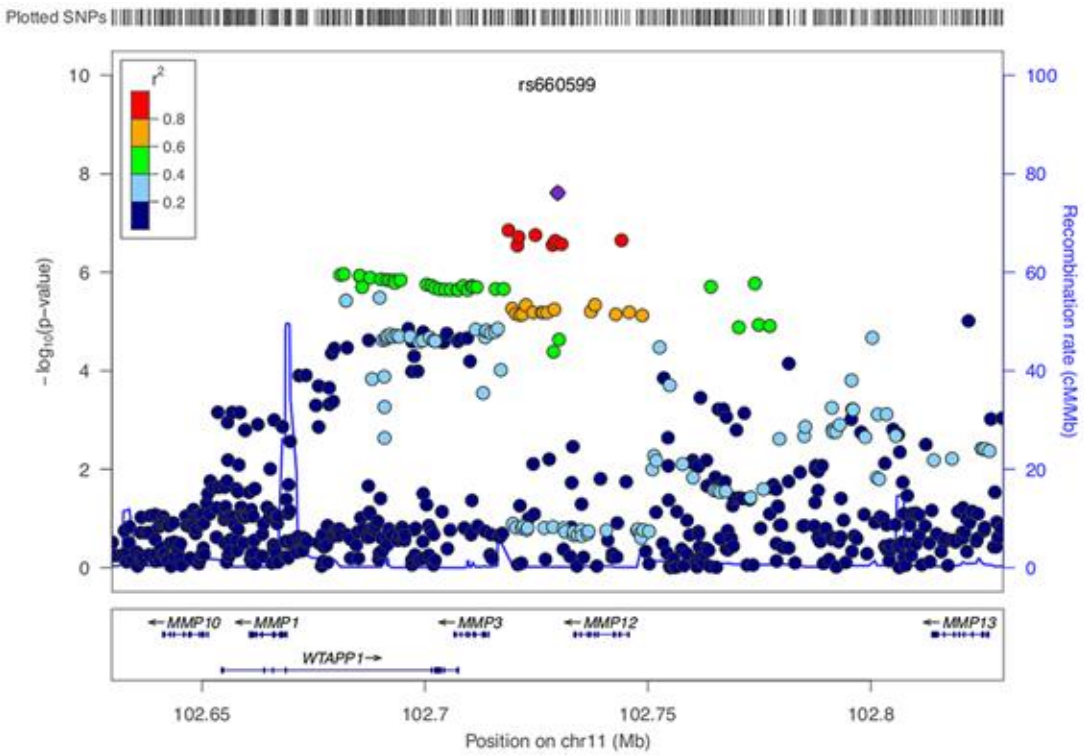
$\beta = 1$
SE $\beta = 0.2$

Z: 1 / 0.20 = **5**

Genotype seems to be predictive of gene expression, very little variability. Very small p-value

Finding expression quantitative trait loci (eQTLs)

- Usually, for every gene SNPs within 1MB of the TSS of the respective gene are tested. (Why?)
- Because many SNPs are correlated with each other, a gene is often associated with a whole LD-block.
- Before multiple testing, “independent signals” are identified using clumping.
- So instead of correcting for the total number of SNPs, you correct for the total number of “independent signals”
- More complex fine mapping methods exist (e.g. SuSiE, Zhang et al. PLoS Genetics 2024)



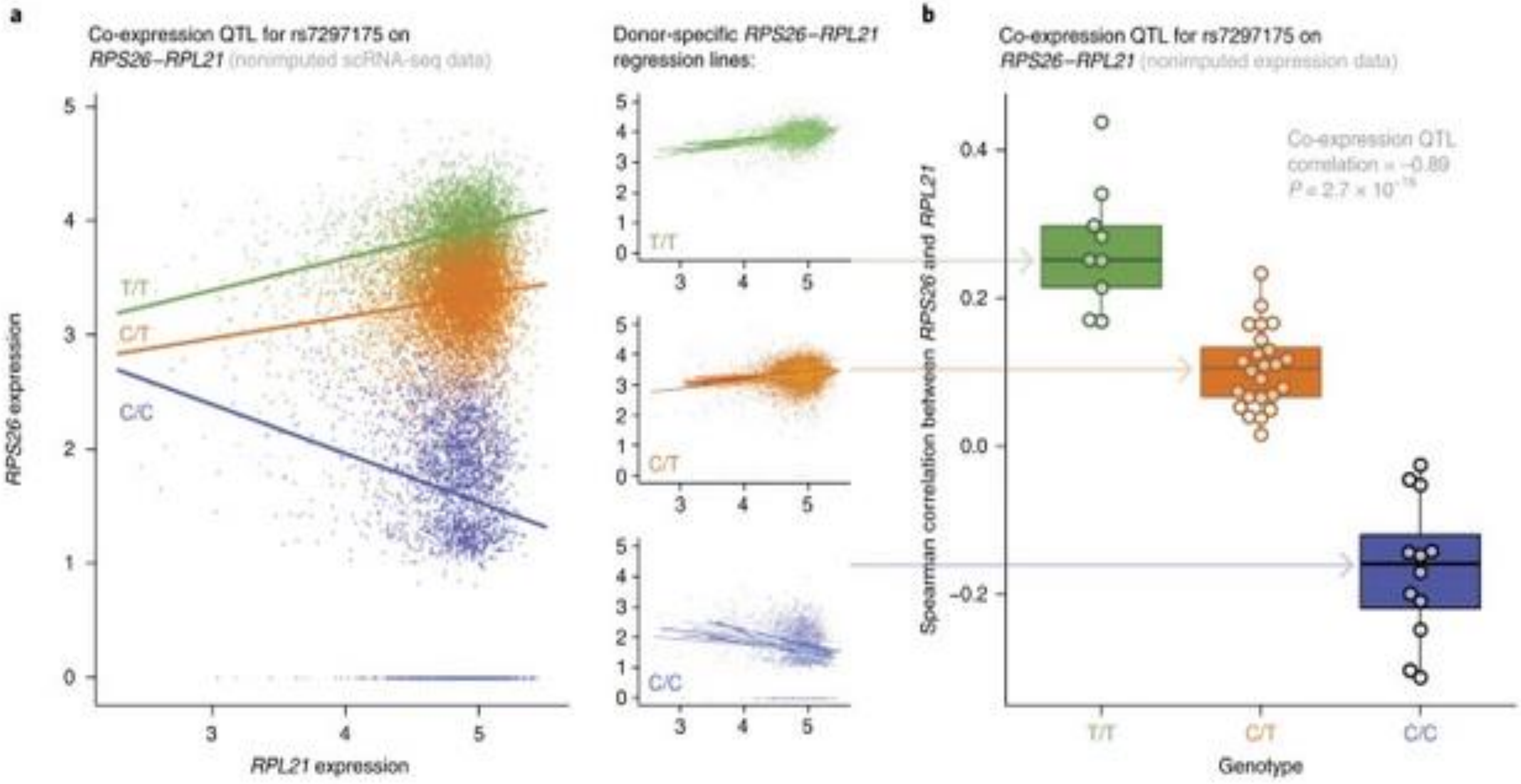
Co-expression QTLs

Letter | Published: 02 April 2018

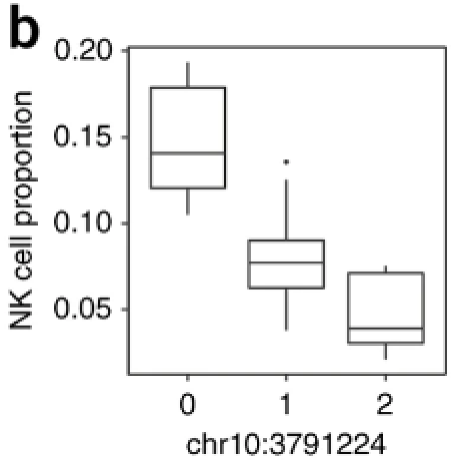
Single-cell RNA sequencing identifies celltype-specific cis-eQTLs and co-expression QTLs

Monique G. P. van der Wijst, Harm Brugge, Dylan H. de Vries, Patrick Deelen, Morris A. Swertz, LifeLines Cohort Study, BIOS Consortium & Lude Franke

Nature Genetics **50**, 493–497 (2018) | Cite this article



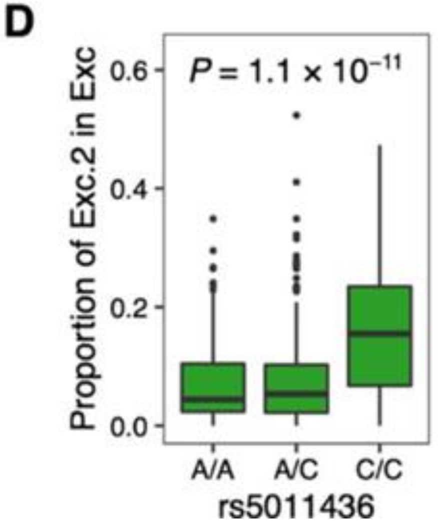
Cell state abundance quantitative trait loci (csaQTLs)



Published: 11 December 2017

Multiplexed droplet single-cell RNA-sequencing using natural genetic variation

Hyun Min Kang, Meena Subramaniam, Sasha Targ, Michelle Nguyen, Lenka Maliskova, Elizabeth McCarthy, Eunice Wan, Simon Wong, Lauren Byrnes, Cristina M Lanata, Rachel E Gate, Sara Mostafavi, Alexander Marson, Noah Zaitlen, Lindsey A Criswell & Chun Jimmie Ye

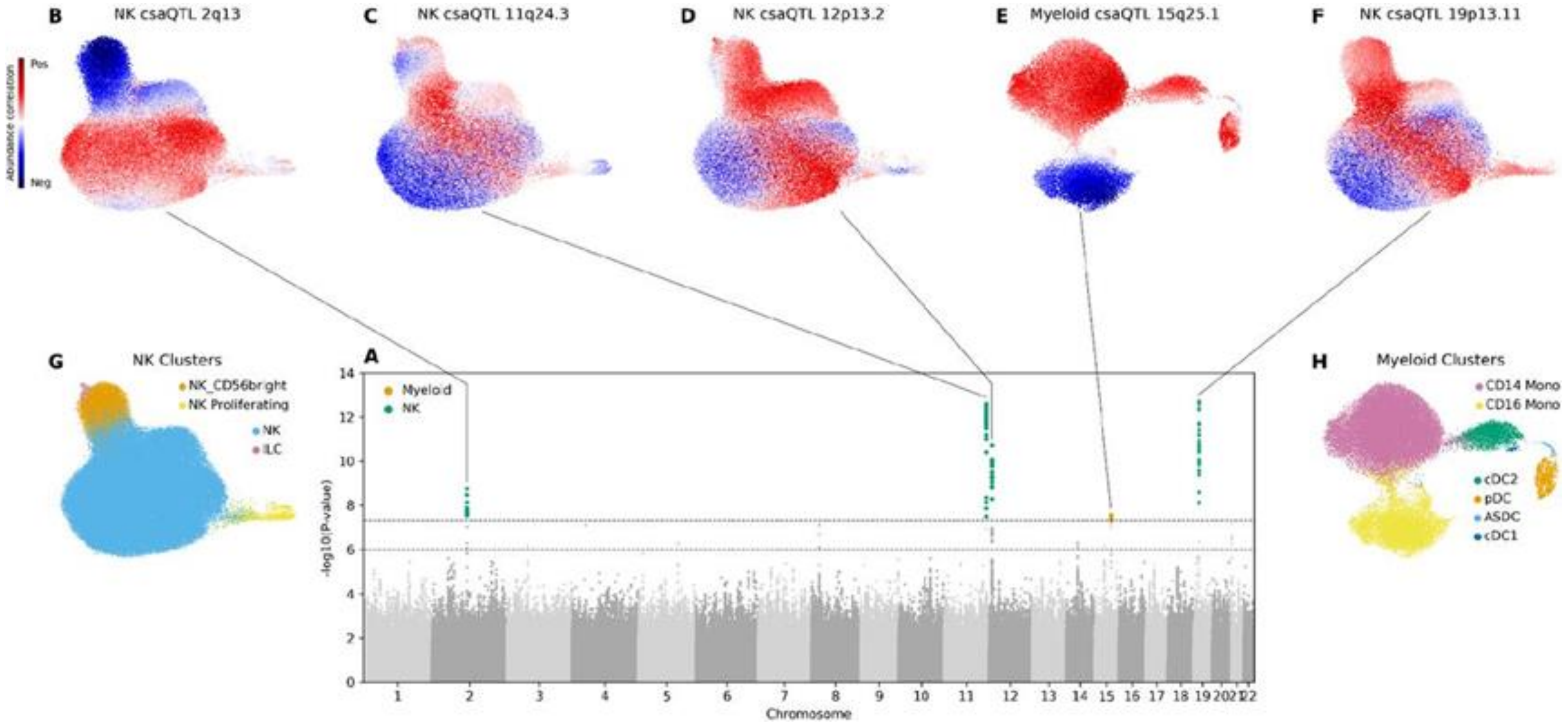


Cell-subtype specific effects of genetic variation in the aging and Alzheimer cortex

Masashi Fujita, Zongmei Gao, Lu Zeng, Cristin McCabe, Charles C. White, Bernard Ng, Gilad Sahar Green, Orit Rozenblatt-Rosen, Devan Phillips, Liat Amir-Zilberstein, Hyo Lee, Richard V. Pearse II, Atlas Khan, Badri N. Vardarajan, Krzysztof Kiryluk, Chun Jimmie Ye, Hans-Ulrich Klein, Gao Wang, Aviv Regev, Naomi Habib, Julie A. Schneider, Yanling Wang, Tracy Young-Pearse, Sara Mostafavi, David A. Bennett, Vilas Menon, Philip L. De Jager

doi: https://doi.org/10.1101/2022.11.07.515446

Cell state abundance quantitative trait loci (csaQTLs)

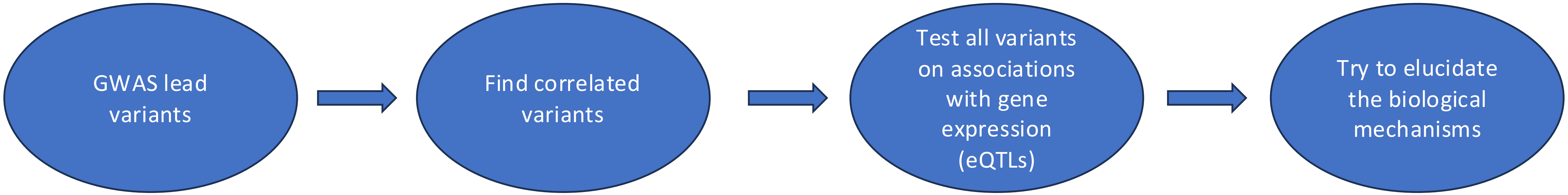


Identifying genetic variants that influence the abundance of cell states in single-cell data

Laurie Rumker, Saori Sakaue, Yakir Reshef, Joyce B. Kang, Seyhan Yazar, Jose Alquicira-Hernandez, Cristian Valencia, Kaitlyn A Lagattuta, Annelise Mah-Som, Aparna Nathan, Joseph E. Powell, Po-Ru Loh, Soumya Raychaudhuri

doi: https://doi.org/10.1101/2023.11.13.566919

Combining GWASs with eQTLs



GWAS papers / GWAS catalog

Based on reference panels (e.g. 1000 genomes, HRC)

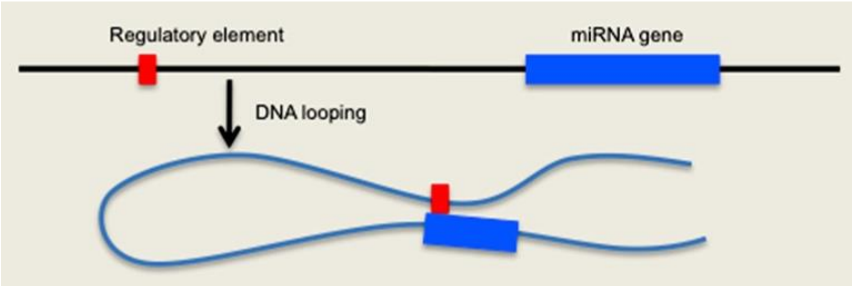
GTEx / your own data

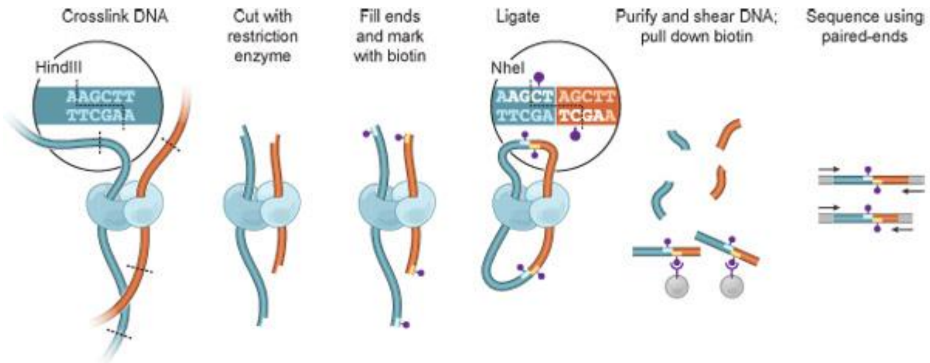
Consequence types (CADD, ensembl)

- 3'-UTR variants
- TFBS variants
- Splice variants
- Methylation
- Histone modifications
- etc

Chromatin interaction

- Identifying regions of DNA that physically interact with each other
- Interaction between distal regulatory elements with promoters to regulate gene expression
- Process:
    - Formaldehyde to covalently link DNA regions that are in close spatial proximity in nucleus
    - DNA is cleaved by restriction digestion and DNA ends are filled in with biotinylated nucleotides.
    - DNA is then ligated to form hybrid DNAmolecules, each corresponding to an interaction event of a pair of loci.
    - Fragmented and sequenced, then mapped to genome

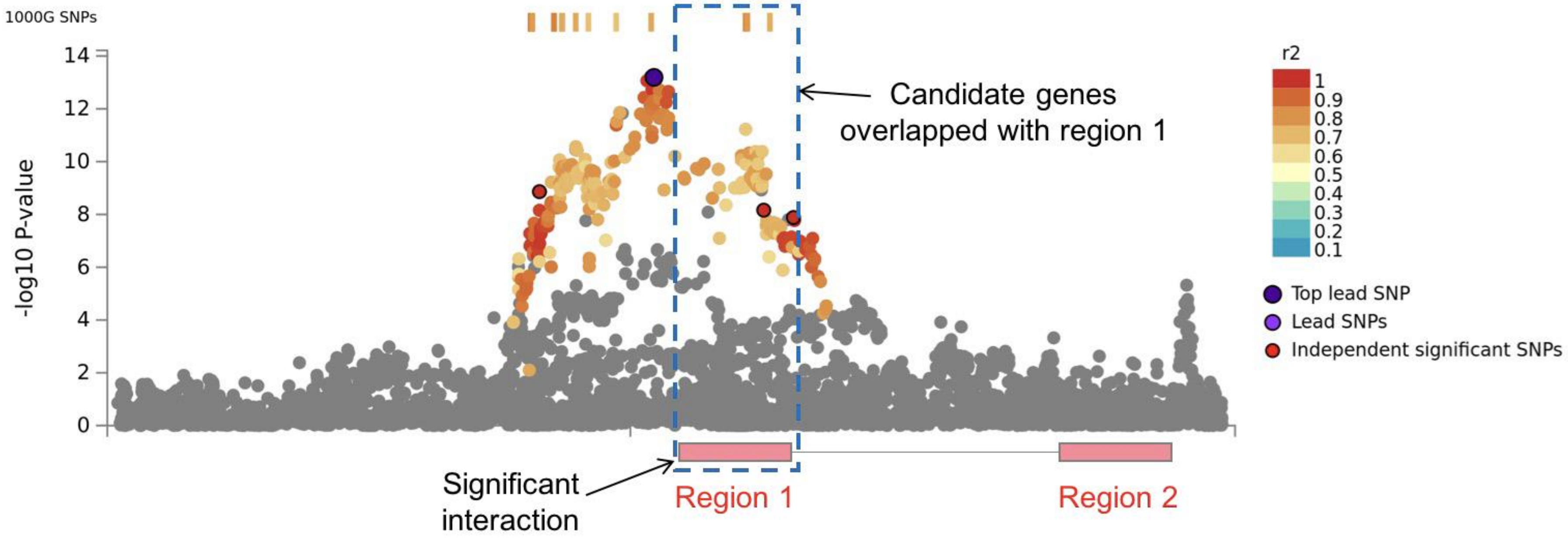




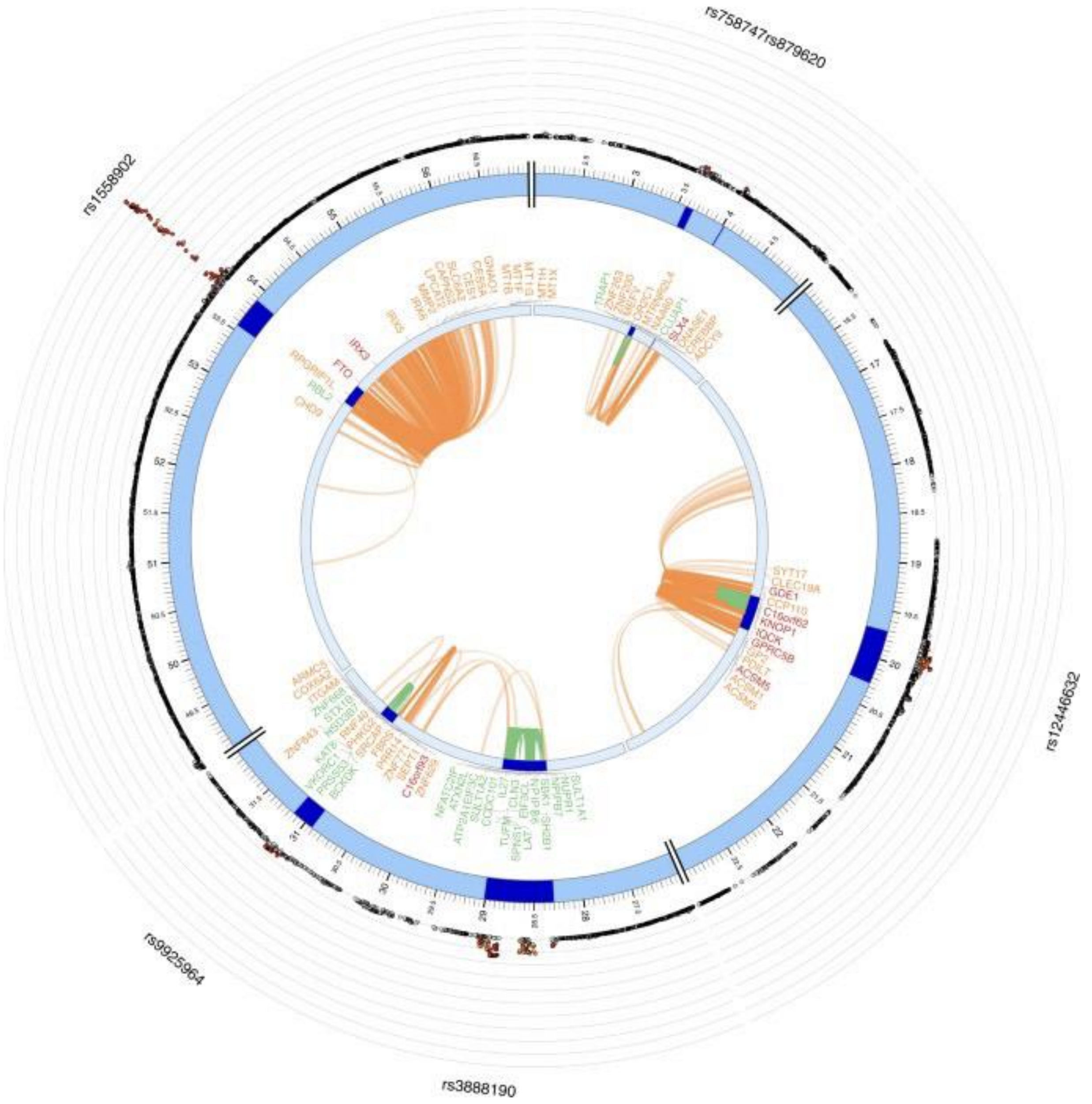
Berkum et al. 2010

Chromatin interaction

- **Region 1:** One end of the interaction that overlaps with one of the candidate SNPs
- **Region 2:** Other end of the significant interaction. Identifies genes whose promoter region interacts with the region containing the candidate SNPs

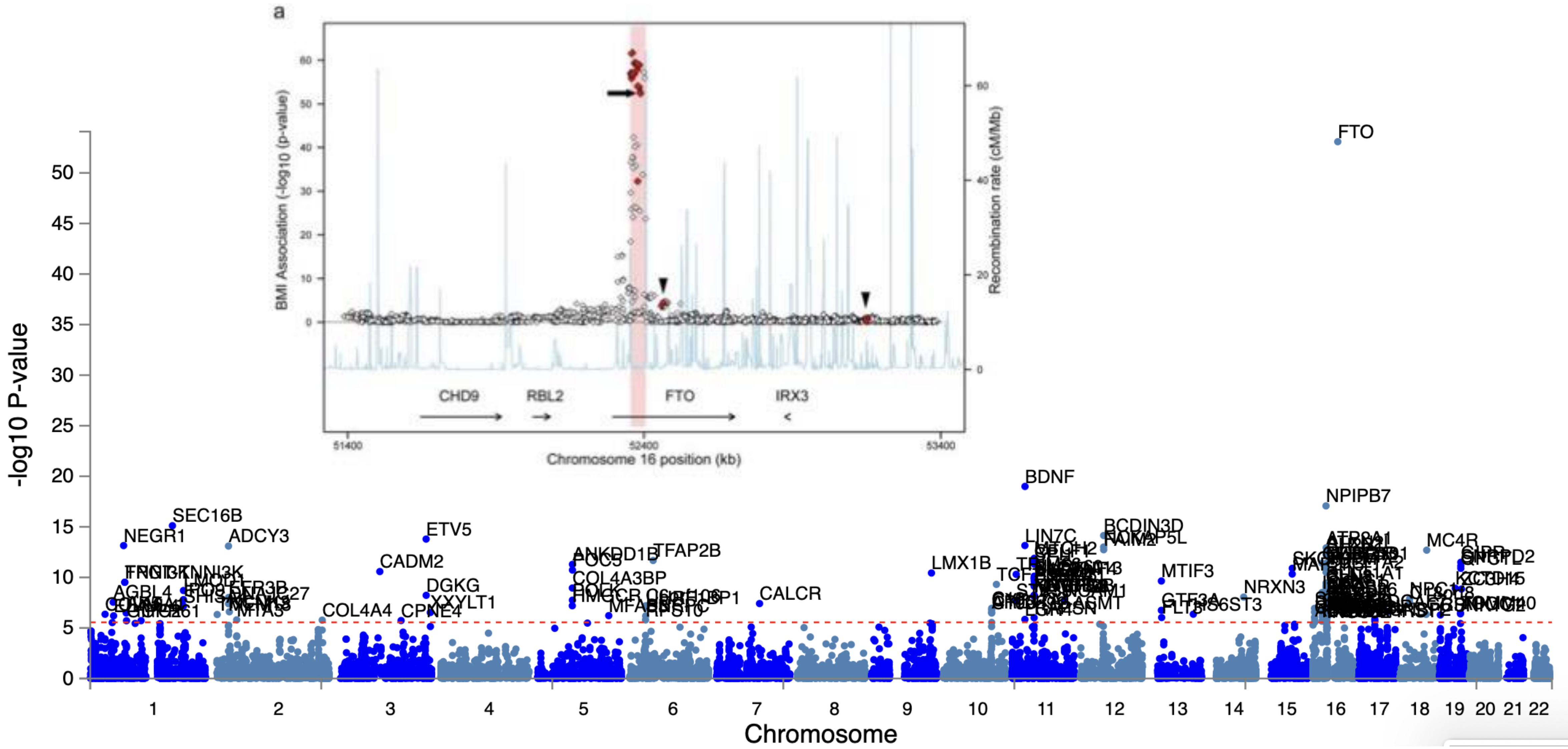


Chromatin interactions and eQTLS of a BMI risk locus on chr16



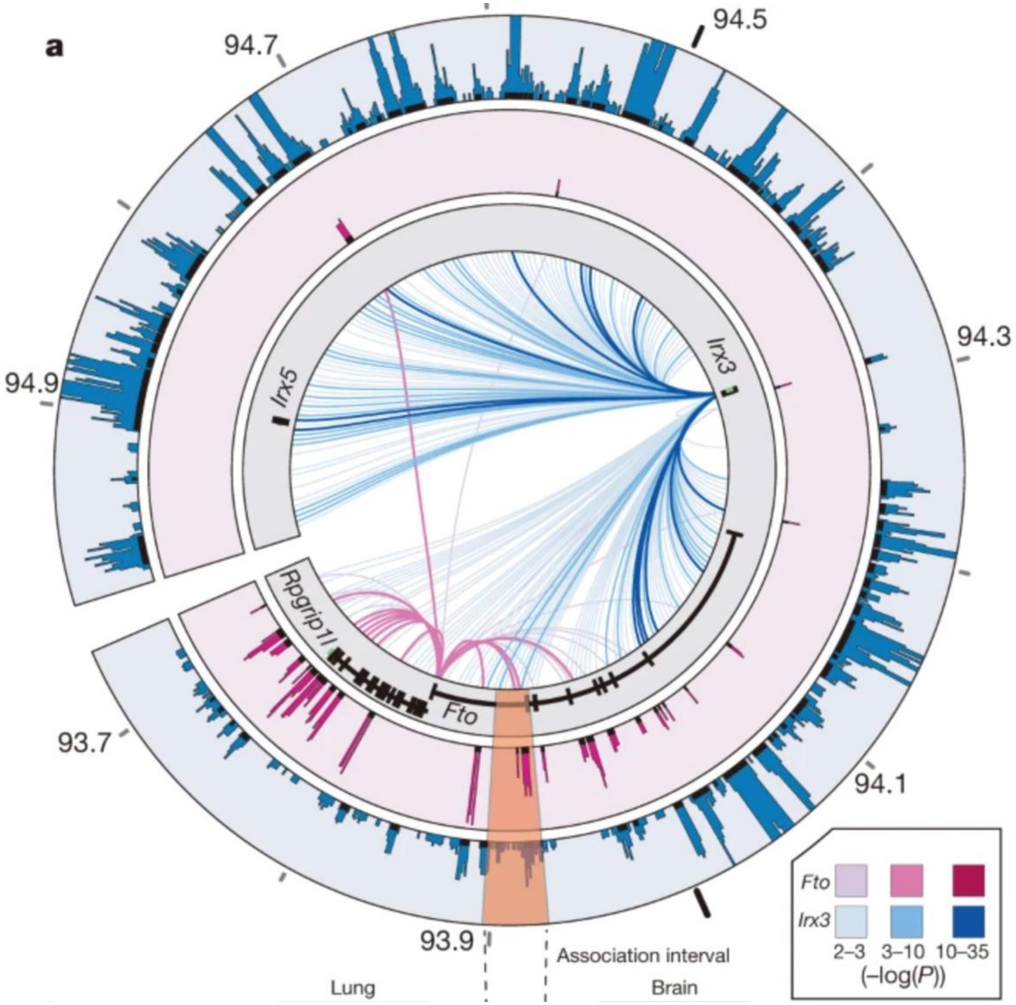
Genes
Orange: mapped by eQTL data
Green: mapped by HiC data
Red: mapped by both

GWAS to mechanism example - FTO



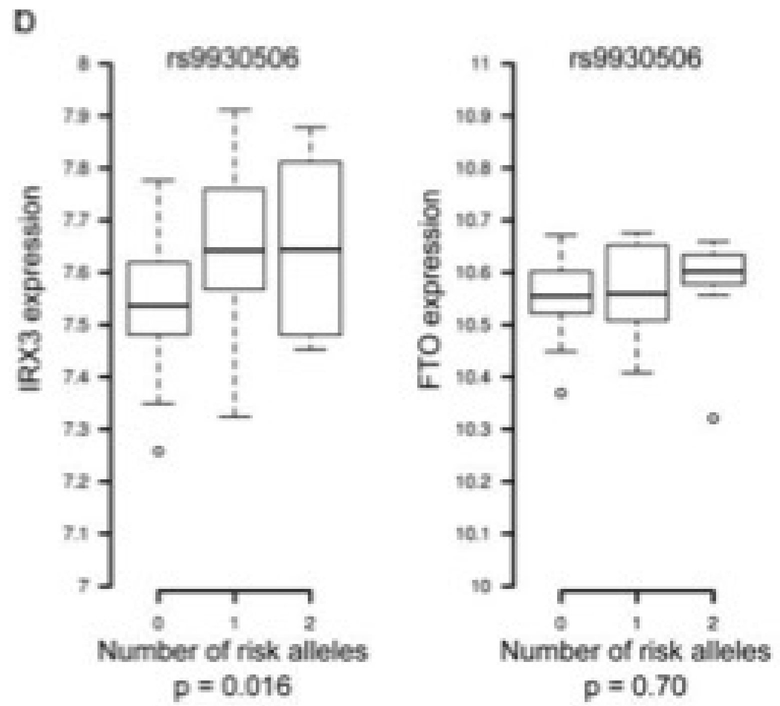
GWAS to mechanism example - FTO

- *The Fto promoter chiefly participates in genomic interactions proximal to the gene promoter.*
- *Interaction between Fto promoter and GWAS region in mouse embryos but not in adult mouse brains*
- *Promoter of Irx3 participates in numerous long-range interactions, including with the GWAS region in both mouse embryo and adult mouse brain, as well as MCF-7 cells and zebrafish embryos*
- *Obesity-associated variants within FTO form long-range functional connections with IRX3*

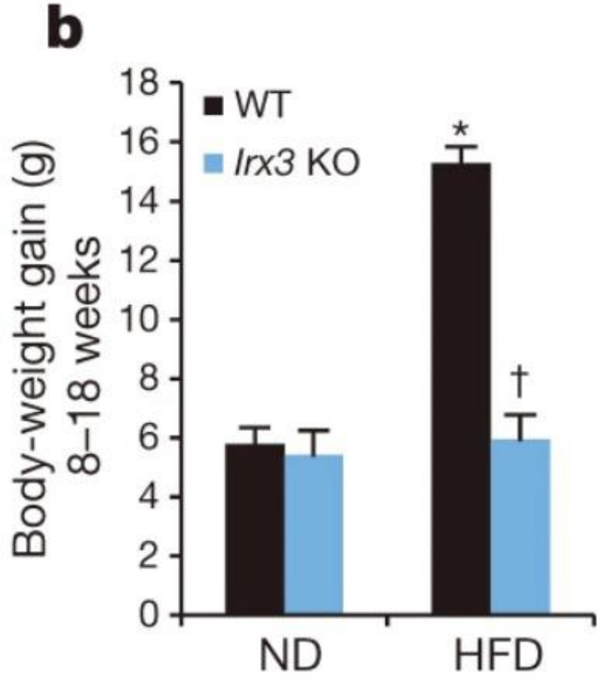


Smemo et al Nature 2014

GWAS to mechanism example - FTO



BMI-associated SNPs are eSNPs for *IRX3*, not *FTO*, expression in human brain



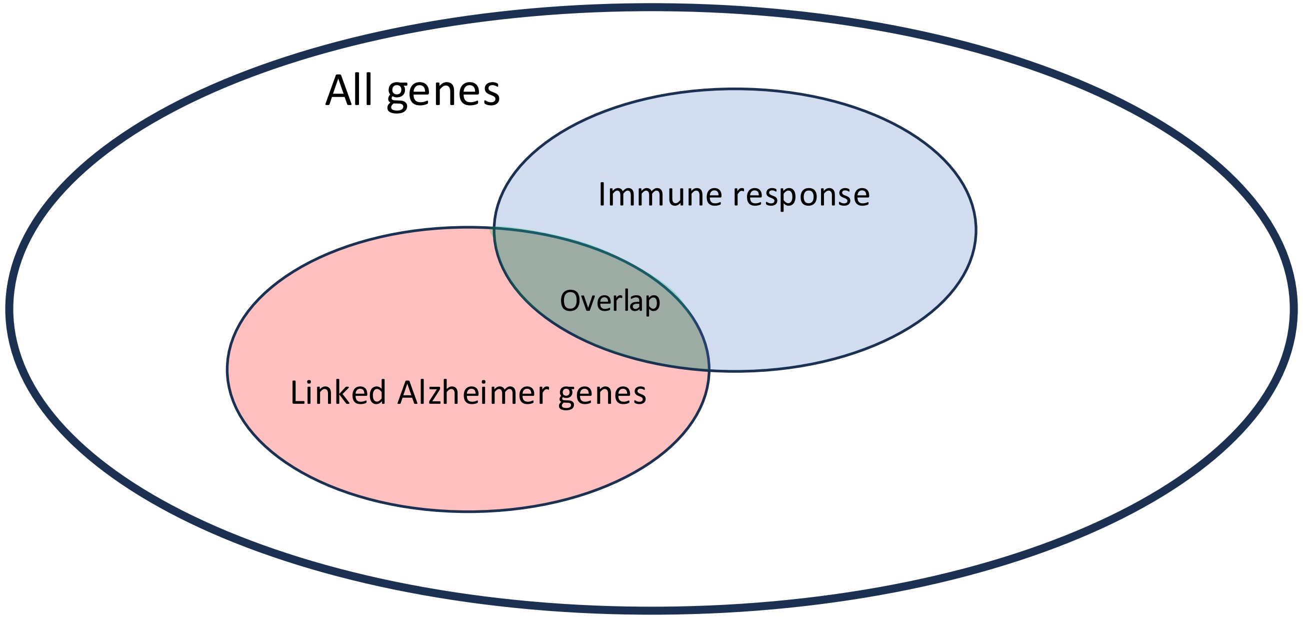
Irx3-deficient mice are leaner and are protected against dietinduced obesity

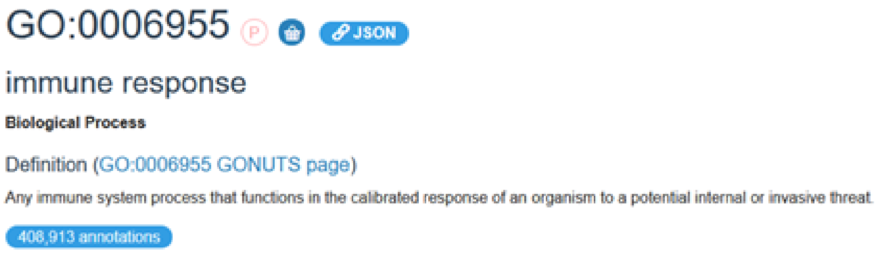
Gene to function

Gene set enrichment analysis

- Assume now that we linked all Alzheimer’s variants to genes.
- What could be next thing that we would want to investigate?
- We can test whether the genes are involved in the same pathway or active in the same biological process, molecular function, cellular component or are they interacting with the same metabolite, and many more!

Gene set enrichment analysis




GO:0006955

immune response

Biological Process

Definition (GO:0006955 GONUTS page)

Any immune system process that functions in the calibrated response of an organism to a potential internal or invasive threat.

408,913 annotations

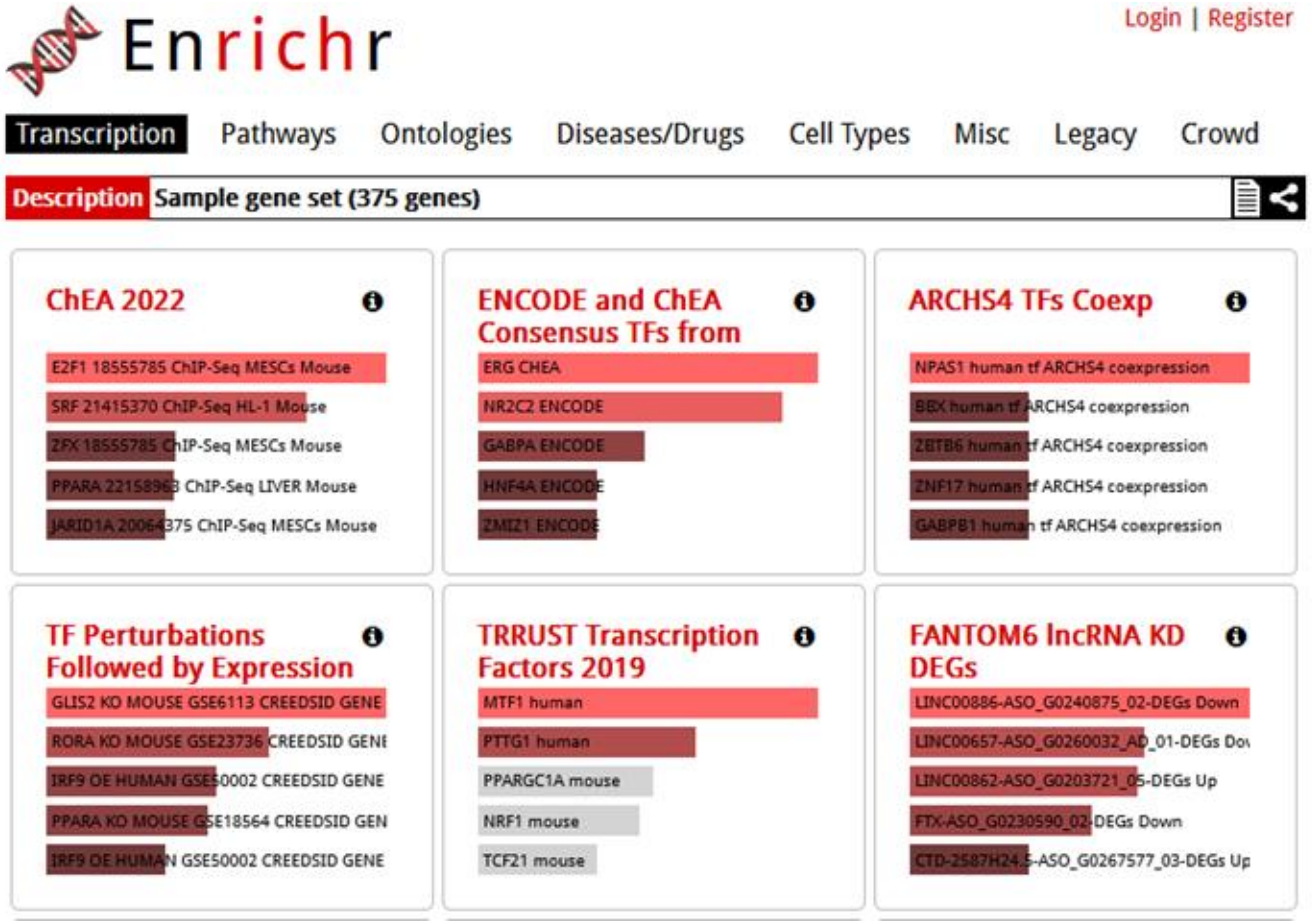
https://www.ebi.ac.uk/QuickGO/term/GO:0006955

Observed	Not Alzheimer Genes (N = 50)	Alzheimer Genes (N = 40)
Not in immune (n =50)	40	10
In Immune (n = 40)	10	30

Expected	Not Alzheimer Genes (N = 50)	Alzheimer Genes (N = 40)
Not in immune (n =50)	27.8	22.2
In Immune (n = 40)	22.2	17.8

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Gene set enrichment analysis



225 different libraries

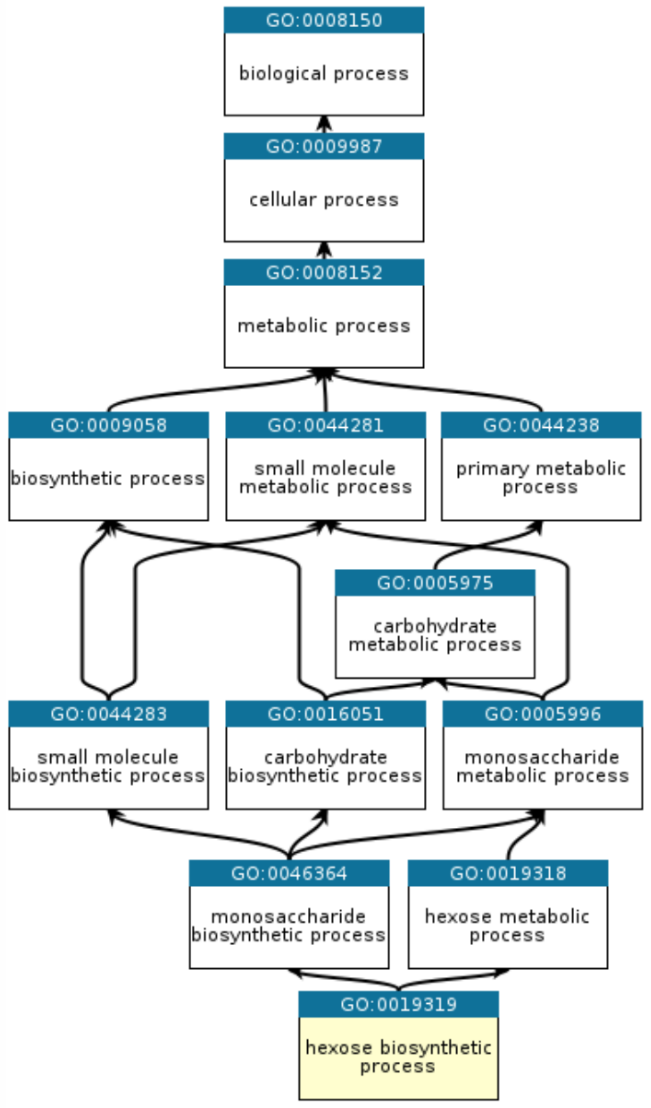
https://maayanlab.cloud/Enrichr/

Gene ontology

- Hierarchy of terms
  - Biological processes
  - Cellular Component
  - Molecular function

- Very extensive (~7.028 terms)

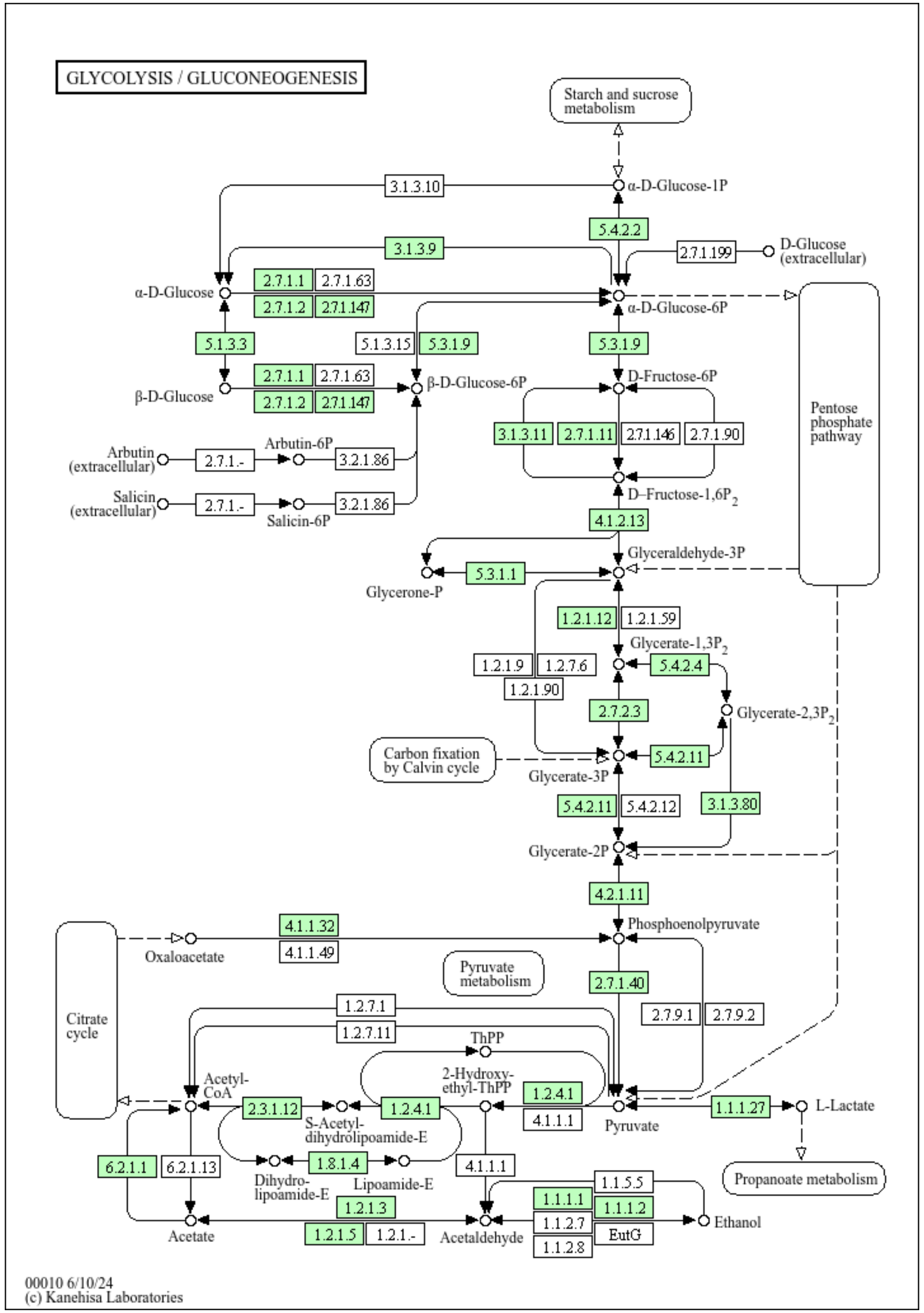
- But a lot of overlap between terms



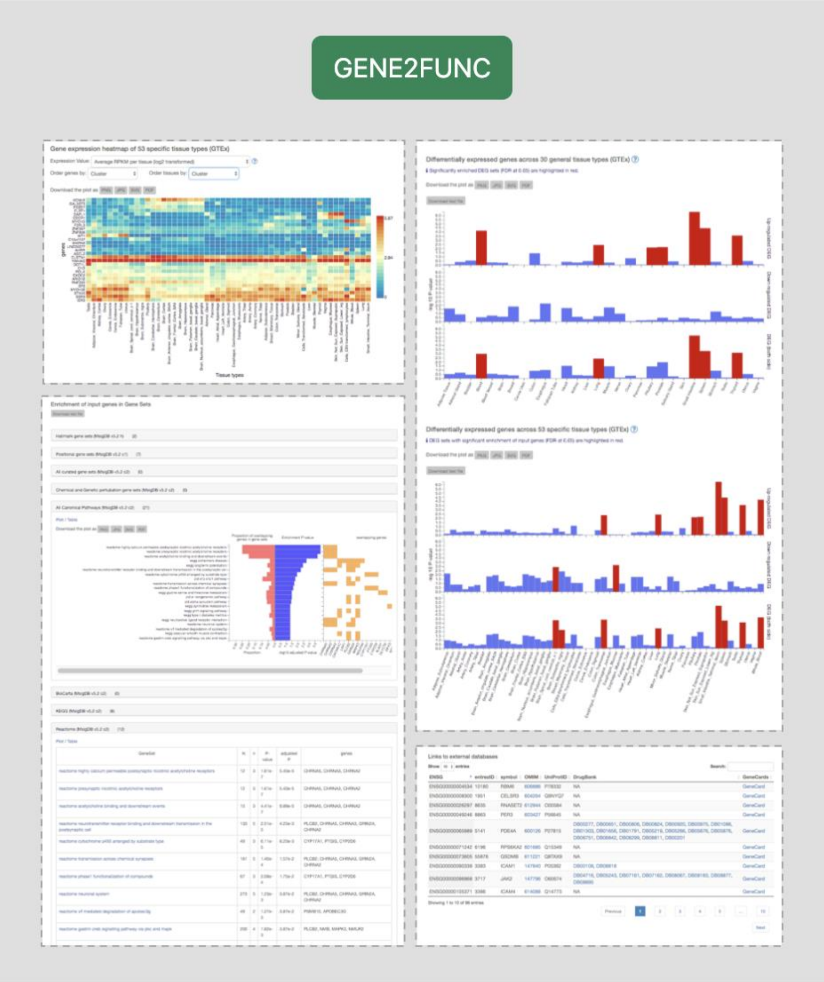
https://geneontology.org/

KEGG Pathway Database

- "KEGG PATHWAY is a collection of manually drawn pathway maps representing our knowledge of the molecular interaction, reaction and relation networks for:"
  - Metabolism
  - Genetic Information Processing
  - Environmental Information Processing
  - Cellular Processes
  - Organismal Systems
  - Human Diseases
  - Drug Development



FUMA GWAS: Functional Mapping and Annotation of Genome-Wide Association Studies



https://fuma.ctglab.nl/

Summary

- Identifying genetic variants using GWAS
- Biological mechanisms of non-coding variants
- Mapping variants to genes (QTL analysis)
- Linking genes to function

Thank you!

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